

Reinforcement Learning — Assignment 1

Unit I: Foundations & Bandit Algorithms | M.Tech AI & DS | VNR VJIET

Name: _____ Roll No.: _____

Q1. Two parents teach a child to ride a bike. Parent A shouts “lean left NOW!” at every moment. Parent B says “good” or “bad” after each attempt and lets the child figure it out. Which child has to *explore*, and what does this say about why RL agents must explore at all?

Q2. Let $A \in \mathbb{R}^{2 \times 3}$, $B \in \mathbb{R}^{3 \times 2}$, $C \in \mathbb{R}^{2 \times 2}$. For each of the following, state whether it is defined and give its shape if so: (a) AB , (b) BA , (c) $AB - C$, (d) $A + B^\top$, (e) $A^\top BA$.

Q3. A classmate looks at a normal distribution plot, points to the peak, and announces: “ $p(0) = 0.4$, so there’s a 40% chance the value is exactly 0.” What two ideas are they confusing, and what does $p(0) = 0.4$ actually mean?

Q4. A discrete random variable X takes values $\{1, 3, 5\}$ with probabilities $\{\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\}$. Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, $\text{Var}(X)$, and the standard deviation σ .

Q5. You order from the same restaurant every day; your friend tries a new one every weekend and complains that half her meals are bad. After a year, which of you is more likely to have actually found the best restaurant in your city, and what is the price each of you paid to get there?

Q6. An ε -greedy agent has $k = 4$ arms and $\varepsilon = 0.4$. (a) What is the probability that the *greedy* arm is selected on a given step? (b) What is the probability that a *specific non-greedy* arm is selected?

Q7. After 9 pulls of an arm, $Q_{10} = 5$. The 10th pull returns $R_{10} = 15$. Compute Q_{11} using (a) the sample-average rule, (b) a constant step size $\alpha = 0.2$. Which update reacts more strongly, and when might you prefer it?

Q8. A startup wants to A/B test 8 banner-ad variants with click rates around 2%. Their CTO suggests plain UCB. What specific feature of this problem makes plain UCB a suboptimal choice, and which algorithm would you reach for instead?

Q9. Two interviewers compare exploration strategies on a 5-armed bandit. Interviewer A uses ε -greedy with $\varepsilon = 0.1$ forever. Interviewer B uses Thompson Sampling. After a million pulls, one of them is still wasting pulls on a clearly bad arm. Which one, and why does the other naturally stop?

Q10. An arm starts with a $\text{Beta}(2, 2)$ prior. (a) After 6 trials yielding 4 successes and 2 failures, state the posterior parameters (α, β) and the posterior mean. (b) Then 4 more trials yield 1 success and 3 failures. State the updated (α, β) and posterior mean.